

Model-Based Actor–Critic Learning of Robotic Impedance Control in Complex Interactive Environment

Xingwei Zhao¹, Shibo Han, Bo Tao², *Member, IEEE*, Zhouping Yin³, *Member, IEEE*, and Han Ding, *Senior Member, IEEE*

Abstract—In complex robot applications, such as human–robot interaction and robot machining, robots should interact with an unknown environment. To learn the interactive skill, a model-based actor–critic learning algorithm and a safety-learning strategy are proposed in this article to find the optimal impedance control, in which the learning process is safe and fully automatic and does not know the system parameter. In the learning algorithm, a critic is defined as a quadratic form of the system states and the external force. A modified deterministic policy gradient algorithm is presented to improve the learning efficiency. The proposed approach utilizes a model-based constraint and a highly efficient learning algorithm. In the safety-learning strategy, the robot is trained under a constant force, and the learned impedance control can transfer to different interaction situations by choosing the suitable impedance index. The effectiveness of the learning algorithm and the performance of the learned impedance control are validated in a UR5 robot. The robot can perform human–robot interaction and robot machining tasks after the training process with 100 s training time.

Index Terms—Actor–critic learning, human–robot interaction, impedance control, robot machining.

I. INTRODUCTION

WITH the increasing need for safe, reliable, and efficient physical robot–environment interaction, impedance control is a permission tool to deal with different interaction tasks, such as robot assembly tasks [1], robot machining [2], human–robot interaction [3], collaborative manufacturing [4], robotic surgical procedures [5], and robotic manipulation [6].

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The authors are with the State Key Laboratory of Digital Manufacturing Equipment and Technology School of Mechanical Science and Engineering, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: zhaoxingwei@hust.edu.cn; m201870633@hust.edu.cn; taobo@hust.edu.cn; yinzhp@hust.edu.cn; dinghan@hust.edu.cn).

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Impedance control is designed to provide stable tracking by regulating the impedance response of a robot when the robot contacts with the external environment [7]. It is usually prescribed to guarantee the safety of interaction with environments [8] or act as an interface for a human to tutor the motion of the robot [1].

The main difficulty in the impedance control of robots is to overcome the confliction between motion accuracy and interaction safety [9]. For example, robots should exhibit a low mechanical impedance, such that they can interact with humans safely. Meanwhile, the robots should reject disturbances, which results in an increased impedance. One way to solve this confliction is to use adaptive impedance control for the time-varying environment [10]. Braun *et al.* [11] proposed variable stiffness control for explosive movement tasks. Kronander *et al.* [12] analyzed the stability properties of a system with variable impedance control. A cascade-loop physical human–robot interaction controller, which can adapt to different users by predicting human motion intent and estimate a reference trajectory, was proposed by Cremer *et al.* [9]. Li *et al.* [13] proposed impedance learning for robots interacting with unknown environments, which aims to adjust impedance parameters to achieve safe and stable contact with humans. They also developed a robot controller that concurrently adapts to feedforward force, impedance, and reference trajectory when a robot interacts with an unknown environment [14]. Xu *et al.* [15] proposed an adaptive impedance control with reference trajectory learning, in which control is designed to follow the target impedance model. The adaptive policy is a powerful tool to balance motion accuracy and interaction safety but is not always suitable. For example, when the external environment changes rapidly or the interaction force may damage the robot or environment, the learning strategy while working may not be suitable.

As an alternative to adaptive control, multiobject controller can be designed to solve the confliction issue. Dietrich *et al.* [16] provided a hierarchical multiobjective controller for trajectory tracking that ensures the asymptotic stability of an equilibrium and the desired contact impedance at the same time. A multiobject is used to overcome the confliction issue, but the system parameters should be given previously for the control design. Reinforcement learning (RL) is a machine learning method used to find an optimal policy based on the given cost

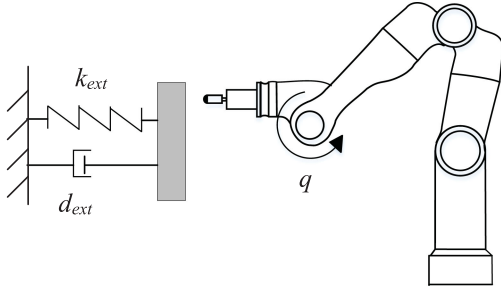


Fig. 1. Impedance control of robots.

function, where the control tasks are formulated as a sequential decision-making process [17]. Thus, the cost function can be used to express different objects, and the confliction issue is transferred to find the suitable cost function with respect to different tasks. RL is widely used for robotic control problems, such as locomotion [18], human–robot interaction [19], stable flight of an autonomous helicopter [20], ball-catching tasks [21], and single-leg hopping [22]. Although RL has been effectively applied to solve various problems, it is rarely used to deal with impedance control. Oikawa *et al.* [23] studied the RL for robotic assembly, where RL is used for the selection of the stiffness matrix. Buchli *et al.* [24] proposed the RL approach for impedance control, where the control includes position and speed feedback but without force feedback. Li *et al.* [25] used integral RL to optimize the parameters of the impedance model, but the system parameters should be partially known. A parameter-free and efficient RL for impedance control of robots should be further investigated. Two main issues arise for the RL of impedance control. First, an exploration process is required for RL, which may introduce the safety issue in the interaction environment. Then, the greedy policy improvement is mainly used in RL, which may result in a low learning speed. The greedy policy is problematic in continuous action spaces because it requires global maximization at every step [26].

To realize RL of robotic impedance control, a model-based actor–critic (AC) learning algorithm and a safety-learning strategy are proposed in this article. The optimal impedance control of robots can be automatically learned without knowing the system and environment parameters. The innovations of this article are as follows.

- 1) Robot point stabilization control and impedance control can be learned synchronously by combining robotic and interaction dynamics into an augmented system. The impedance property and stability region can be adjusted separately by introducing an impedance index to overcome the confliction between motion accuracy and interaction safety.
- 2) A modified deterministic policy gradient algorithm based on [27] is proposed, in which the model-based feature is integrated into the policy update law. The learning efficiency can be ensured.
- 3) A safety-learning strategy is proposed to train the optimal impedance control. For safety reasons, a constant force is provided to the system during the training process. A

way to transfer the learned results to the human–robot interaction and robot machining situations is discussed.

The rest of the article is organized as follows. Section II defines the optimal impedance control. In Section III, a model-based AC learning algorithm for impedance control is proposed. A safety-learning strategy based on model-based AC learning is investigated in Section IV. Sections V and VI show the simulation and experimental results, respectively. Finally, Section VII concludes this article.

II. INTERPRETATION OF OPTIMAL IMPEDANCE CONTROL

Impedance control consists of state and force feedback. The robot under impedance control can return the null position in free space and acts as a spring when interacting with external obstacles.

Given a single degree-of-freedom robot joint, as shown in Fig. 1, the dynamics of the robotic system can be described as a second-order function as

$$M\ddot{q} + D\dot{q} + mgl \sin(q) = u + af \quad (1)$$

where q is the joint angle, M is the joint inertia, m is the joint mass, D is the friction coefficient, g is the gravity constant, u is the input torque, f is the external force, a is a constant, and l is the length of the force arm. Linearizing the dynamic equation, we have

$$\dot{\mathbf{X}} = \mathbf{A}_c \mathbf{X} + \mathbf{B}_c u + \mathbf{E}_c f, \quad (2)$$

$$\mathbf{A}_c = \begin{bmatrix} 0 & 1 \\ -M^{-1}mgl \cos q & -M^{-1}D \end{bmatrix},$$

$$\mathbf{B}_c = \begin{bmatrix} 0 \\ M^{-1} \end{bmatrix}, \mathbf{E}_c = \begin{bmatrix} 0 \\ aM^{-1} \end{bmatrix}, \mathbf{X} = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}$$

where $\mathbf{X}_k$ is the robot state, and $\mathbf{A}_c$, $\mathbf{B}_c$, and $\mathbf{E}_c$ are the system matrices. The discretized dynamic equation is

$$\mathbf{X}_{k+1} = \mathbf{A}_d \mathbf{X}_k + \mathbf{B}_d u_k + \mathbf{E}_d f_k, \quad (3)$$

$$\mathbf{X}_k = [q_k \ \dot{q}_k]^T$$

where $\mathbf{X}_k$ is the robot state in discrete form, $\mathbf{A}_d$, $\mathbf{B}_d$, and $\mathbf{E}_d$ are the discrete forms of $\mathbf{A}_c$, $\mathbf{B}_c$, and $\mathbf{E}_c$, respectively.

Impedance control based on the discretized dynamic equation is expressed as

$$u_k = \mathbf{K}_x \mathbf{X}_k + K_f f_k \quad (4)$$

where $\mathbf{K}_x$ and K_f are the control feedback gains of the state and the external force, respectively. To make the system stable, a way to design the control law is expressed as follows.

The external force can generally be described as

$$f_{k+1} = \mathbf{A}_{fx} \mathbf{X}_k + A_{ff} f_k \quad (5)$$

where $\mathbf{A}_{fx}$ indicates the part in which the external force relates to the system state, and A_{ff} indicates the part in which the external force relates to itself. Equations (3) and (5) can constitute an augmented system as

$$\bar{\mathbf{X}}_{k+1} = \bar{\mathbf{A}} \bar{\mathbf{X}}_k + \bar{\mathbf{B}} u_k, \quad (6)$$

$$\bar{\mathbf{A}} = \begin{bmatrix} \mathbf{A}_d & \mathbf{E}_d \\ \mathbf{A}_{fx} & A_{ff} \end{bmatrix}, \bar{\mathbf{B}} = \begin{bmatrix} \mathbf{B}_d \\ 0 \end{bmatrix}, \bar{\mathbf{X}}_k = \begin{bmatrix} \mathbf{X}_k \\ f_k \end{bmatrix}$$

where $\bar{\mathbf{X}}_k$ is the augmented state including the states and the external force, and $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ are the augmented system matrix.

As designers, we want the system to find a balance among tracking behavior, impedance property, and input energy. Then, the cost function is defined as

$$c_k = \frac{1}{2} \left((q_k + Q_f f_k)^T (q_k + Q_f f_k) + \dot{q}_k^T Q_d \dot{q}_k + u_k^T R u_k \right) \quad (7)$$

where Q_d , Q_f , and R are the weight factors. Considering that factors Q_d and Q_f are related to the impedance property, we name them impedance index. The cost function can be written in a compacted form based on the augmented state as

$$c_k = \frac{1}{2} \bar{\mathbf{X}}_k^T \bar{\mathbf{Q}} \bar{\mathbf{X}}_k + \frac{1}{2} u_k^T R u_k, \quad (8)$$

$$\bar{\mathbf{Q}} = \begin{bmatrix} 1 & 0 & Q_f \\ 0 & Q_d & 0 \\ Q_f & 0 & Q_f^2 \end{bmatrix}$$

where $\bar{\mathbf{Q}}$ is the matrix form of the impedance index, which is a weight matrix of the variables $\bar{\mathbf{X}}_k$. The optimal impedance control can be interpreted as the optimal control of the augmented system (6) with the impedance index (8).

In general, the optimal stable impedance control law can be obtained by solving the algebraic Riccati equation (ARE)

$$u_k = \mathbf{K}_x \mathbf{X}_k + K_f f_k = \mathbf{K} \bar{\mathbf{X}}_k, \quad (9)$$

$$\mathbf{K} = -(\mathbf{R} + \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{B}})^{-1} \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{A}}$$

where $\mathbf{K} = [\mathbf{K}_x \ K_f]$, and $\mathbf{P}$ satisfies the following ARE:

$$\bar{\mathbf{Q}} - \mathbf{P} + \bar{\mathbf{A}}^T \mathbf{P} \bar{\mathbf{A}} - \bar{\mathbf{A}}^T \mathbf{P} \bar{\mathbf{B}} (\mathbf{R} + \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{B}})^{-1} \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{A}} = 0. \quad (10)$$

The optimal impedance control law is related to the impedance index $\bar{\mathbf{Q}}$.

To better choose the impedance index $\bar{\mathbf{Q}}$, its physical meaning, i.e., the connection between the index and the impedance capability, should be interpreted.

In accordance with (7), we have

$$c_k \rightarrow \min (q_k + Q_f f_k)^T (q_k + Q_f f_k) + \dot{q}_k^T Q_d \dot{q}_k + u_k^T R u_k \approx 0 \quad (11)$$

and

$$Q_f = -\frac{q_k}{f_k}, \text{ if } Q_d \ll 1 \text{ and } R \ll 1. \quad (12)$$

Thus, Q_f defines the flexibility of the system under impedance control.

On the contrary, if $q_k + Q_f f_k = 0$, we have

$$c_k = \dot{q}_k^T Q_d \dot{q}_k + u_k^T R u_k. \quad (13)$$

Increasing Q_d will increase the weight of the speed, which means increasing the damping of the system. Adjusting R is only necessary when the system has input saturation [28]. Otherwise, R can be a small value.

Remark 1: Even though the impedance index is related to the control feedback $\mathbf{K}$, adjusting the impedance index has several advantages compared to adjusting $\mathbf{K}$. First, the control gain solved by the impedance index can stabilize system (6). In

addition, the impedance index is with clear physical meaning. The designer can easily adjust the impedance index with the physical understanding.

III. MODEL-BASED AC LEARNING FOR OPTIMAL IMPEDANCE CONTROL

Although the optimal impedance control law can be obtained by solving ARE, the system parameters and external stiffness should be known previously. In practice, the system parameters and external stiffness are difficult to measure. To cope with this difficulty, a model-based AC learning algorithm is proposed to solve the optimal impedance control without knowing the system parameters. Then, the impedance control of robots can be learned fully automatically.

To minimize the cost function defined in (8), a value function is defined as the accumulated cost as

$$V(\bar{\mathbf{X}}_k) = \frac{1}{2} \sum_{i=k}^{\infty} \gamma^{i-k} [\bar{\mathbf{X}}_i^T \bar{\mathbf{Q}} \bar{\mathbf{X}}_i + u_i^T R u_i] \quad (14)$$

where $V(\bar{\mathbf{X}}_k)$ is the accumulated cost of c_k from $i = k$ to infinity. Since the accumulated cost may be infinity when $\bar{\mathbf{A}}$ is not Hurwitz, a discount factor $\gamma < 1$ is introduced to make the value function $V(\bar{\mathbf{X}}_k)$ bounded. When $\bar{\mathbf{A}}$ is Hurwitz, $\gamma = 1$.

Summing (14), the value function can be written in a quadratic form as [28]

$$V(\bar{\mathbf{X}}_k) \approx \frac{1}{2} \bar{\mathbf{X}}_k^T \mathbf{P} \bar{\mathbf{X}}_k = \frac{1}{2} (\bar{\mathbf{X}}_k \otimes \bar{\mathbf{X}}_k)^T \text{vec}(\mathbf{P}) \quad (15)$$

where $\otimes$ is the Kronecker product, vec is the vectorization operator, and $\mathbf{P}$ is a positive definite matrix. Based on the value function, the critic function is defined as

$$Q_v(\bar{\mathbf{X}}_k, u_k) = c_k + \gamma V(\bar{\mathbf{X}}_{k+1}). \quad (16)$$

Consequently, the critic function can be written as

$$Q_v(\bar{\mathbf{X}}_k, u_k) = \frac{1}{2} (\varphi_k \otimes \varphi_k)^T \text{vec}(\mathbf{H})$$

$$\text{with } \varphi_k = \begin{bmatrix} \bar{\mathbf{X}}_k \\ u_k \end{bmatrix}, \mathbf{H} = \begin{bmatrix} \mathbf{H}_{xx} & \mathbf{H}_{xu} \\ \mathbf{H}_{ux} & H_{uu} \end{bmatrix} \quad (17)$$

where $\mathbf{H}$ is a weight matrix. The value function defined in (14) can be further written in the form of the Bellman equation as

$$V(\bar{\mathbf{X}}_k) = \frac{1}{2} (\bar{\mathbf{X}}_k^T \bar{\mathbf{Q}} \bar{\mathbf{X}}_k + u_k^T R u_k) + \frac{1}{2} \sum_{i=k+1}^{\infty} \gamma^{i-k} [\bar{\mathbf{X}}_i^T \bar{\mathbf{Q}} \bar{\mathbf{X}}_i + u_i^T R u_i]$$

$$= \frac{1}{2} (\bar{\mathbf{X}}_k^T \bar{\mathbf{Q}} \bar{\mathbf{X}}_k + u_k^T R u_k) + \gamma V(\bar{\mathbf{X}}_{k+1}). \quad (18)$$

Substituting (18)–(16)

$$Q_v(\bar{\mathbf{X}}_k, u_k) - \gamma Q_v(\bar{\mathbf{X}}_{k+1}, u_{k+1}) = c_k + V(\bar{\mathbf{X}}_{k+1}) - \gamma (c_{k+1} + V(\bar{\mathbf{X}}_{k+2})). \quad (19)$$

The critic function is expressed as the Bellman equation

$$Q_v(\bar{\mathbf{X}}_k, u_k) = c_k + \gamma Q_v(\bar{\mathbf{X}}_{k+1}, u_{k+1}) \quad (20)$$

which is the recursive form of the critic function.

From the definition of the critic in (17) and its recursive form (20), the critic evaluation can be performed recursively using the Bellman equation

$$\begin{aligned} & \frac{1}{2}(\varphi_k \otimes \varphi_k - \gamma \varphi_{k+1} \otimes \varphi_{k+1})^T \text{vec}(\mathbf{H}) \\ &= \frac{1}{2}(\bar{\mathbf{X}}_k^T \bar{\mathbf{Q}} \bar{\mathbf{X}}_k + u_k^T R u_k). \end{aligned} \quad (21)$$

Subsequently, the unknown matrix $\mathbf{H}$ can be updated using the recursive least square algorithm as

$$\begin{aligned} \text{vec}(\mathbf{H}^{i+1}) &= \text{vec}(\mathbf{H}^i) + \frac{\mathbf{T}^i \varphi_{e,k}^T (c_k - \varphi_{e,k}^T \text{vec}(\mathbf{H}^i))}{1 + \varphi_{e,k}^T \mathbf{T}^i \varphi_{e,k}}, \\ \mathbf{T}^{i+1} &= \mathbf{T}^i - \frac{\mathbf{T}^i \varphi_{e,k} \varphi_{e,k}^T \mathbf{T}^i}{1 + \varphi_{e,k}^T \mathbf{T}^i \varphi_{e,k}}, \\ \varphi_{e,k} &= \varphi_k \otimes \varphi_k - \gamma \varphi_{k+1} \otimes \varphi_{k+1} \end{aligned} \quad (22)$$

where $\mathbf{T}^i$ is a weight matrix in the i th iteration, which is to adjust the learning speed of the critic. With the critic function to hand, the actor can be directly learned from the critic. The common approach used to determine the actor is to find the maximization of the critic function

$$\mathbf{K} = \arg \max_{u_k} Q_v(\bar{\mathbf{X}}_k, u_k) \quad (23)$$

where $\mathbf{K}$ is the control policy, including state and force feedback. To accelerate the learning speed, a deterministic policy gradient algorithm was proposed in [27], in which the policy was updated in the direction of the critic function gradient

$$\mathbf{K}^{i+1} = \mathbf{K}^i + \alpha E_{\bar{\mathbf{X}}_k} [\nabla_{\mathbf{K}} Q_v(\bar{\mathbf{X}}_k, u_k)] \quad (24)$$

where $\nabla_{\mathbf{K}} Q_v(\bar{\mathbf{X}}_k, u_k)$ is the critic function gradient with respect to the control policy, $E_{\bar{\mathbf{X}}_k}[\nabla_{\mathbf{K}} Q_v(\bar{\mathbf{X}}_k, u_k)]$ is the expectation function of the gradient, and α is the learning rate.

In accordance with the chain rule, the gradient descent of the critic with $\mathbf{K}$ can be expressed as the gradient descent of the critic with u_k multiplied by the gradient descent of u_k with $\mathbf{K}$

$$E_{\bar{\mathbf{X}}_k} [\nabla_{\mathbf{K}} Q_v(\bar{\mathbf{X}}_k, u_k)] = E_{\bar{\mathbf{X}}_k} [\nabla_{u_k} Q_v(\bar{\mathbf{X}}_k, u_k) \nabla_{\mathbf{K}} u_k]. \quad (25)$$

Equations (9) and (17) are substituted into (25), and the model information can be added to the update law

$$E_{\bar{\mathbf{X}}_k} [\nabla_{\mathbf{K}} Q_v(\bar{\mathbf{X}}_k, u_k)] = E_{\bar{\mathbf{X}}_k} [(2\mathbf{H}_{ux} + 2\mathbf{H}_{uu}\mathbf{K}) (\bar{\mathbf{X}}_k \bar{\mathbf{X}}_k^T)]. \quad (26)$$

On the basis of the average gradient update law, the evaluation of the actor can be expressed as

$$\mathbf{K}^{i+1} = \mathbf{K}^i + \frac{2\alpha}{N} \sum_{k=iN+1}^{(i+1)N} [(\mathbf{H}_{ux} + \mathbf{H}_{uu}\mathbf{K}^i) (\bar{\mathbf{X}}_k \bar{\mathbf{X}}_k^T)] \quad (27)$$

where i indicates the i th update of the actor. The update law approaches zero when $\bar{\mathbf{X}}_k \bar{\mathbf{X}}_k^T$ is small. To eliminate the influence of the system state to the update law, $\bar{\mathbf{X}}_k \bar{\mathbf{X}}_k^T$ can be removed

$$\mathbf{K}^{i+1} = \mathbf{K}^i + \frac{2\alpha}{N} \sum_{k=iN+1}^{(i+1)N} (\mathbf{H}_{ux} + \mathbf{H}_{uu}\mathbf{K}^i). \quad (28)$$

Equation (28) provides the modified model-based update law of the actor based on the deterministic policy gradient algorithm. Compared with the update law in (27), the modified update law works even when $\bar{\mathbf{X}}_k \bar{\mathbf{X}}_k^T$ approaches zero. In summary, the AC learning algorithm for impedance control can be written as follows.

Critic update:

The critic update law is given as

$$\nabla \text{vec}(\mathbf{H}^i) = \frac{T^i \varphi_{e,k}^T}{1 + \varphi_{e,k}^T T^i \varphi_{e,k}} \left(c_k - \varphi_{e,k}^T \text{vec} \left(\begin{bmatrix} \mathbf{H}_{xx} & \mathbf{H}_{xu} \\ \mathbf{H}_{ux} & \mathbf{H}_{uu} \end{bmatrix} \right) \right). \quad (29)$$

Actor update:

The evaluation of the actor is expressed as

$$\nabla \mathbf{K}^i = \frac{2\alpha}{N} \sum_{k=1}^N (\mathbf{H}_{ux} + \mathbf{H}_{uu}\mathbf{K}^i). \quad (30)$$

Theorem 1: Given the augmented system (6) with the critic and actor update laws (29) and (30), the learned actor has the form

$$u_k = \mathbf{K} \bar{\mathbf{X}}_k, \mathbf{K} = -(R + \gamma \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{B}})^{-1} \gamma \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{A}} \quad (31)$$

where $\mathbf{K}$ and $\mathbf{P}$ are equivalent to those in (9) and (10) when $\gamma = 1$. The control law $\mathbf{K}$ minimizes the value function (14) over all stabilizing controls. The detailed proof is provided in the Appendix.

Remark 2: The control law obtained by RL is equivalent to the control law solved using ARE. However, the RL algorithm does not require preknown system and environment parameters.

IV. SAFETY-LEARNING STRATEGY FOR IMPEDANCE CONTROL

For RL, exploration probe noise is normally required during training. The probe noise may result in a safety issue when a robot interacts with the environment. Accordingly, a safety-learning strategy is investigated in this part.

The interaction between the robot and environment is complicated, which may lead to different $\mathbf{A}_{fx}$ and $\mathbf{A}_{ff}$. If the dynamics of the external force change, the system should be learned again to adapt to the new environment. This learning process is inconvenient and time-consuming. Also, the new environment could be dangerous which is not suitable for a training process, such as for the robot machining task.

To cope with it, we follow the idea of transfer learning. Although the training environment is not identical to the testing environment, the learned result can be transferred to the testing environment if the impedance control satisfies some conditions. In general, the training environment is $S_1 = (\mathbf{A}_{fx,1}, \mathbf{A}_{ff,1})$, and the testing environment is $S_2 = (\mathbf{A}_{fx,2}, \mathbf{A}_{ff,2})$. The question is how to choose an impedance index, such that the learned result can transfer to the testing environment.

In the training scene, we choose a constant force to train the robot system, considering that a constant force is easy to exert and does not introduce a safety issue. Mathematically, the training environment is

$$\mathbf{A}_{fx} = \mathbf{0}, \mathbf{A}_{ff} = \mathbf{I}. \quad (32)$$

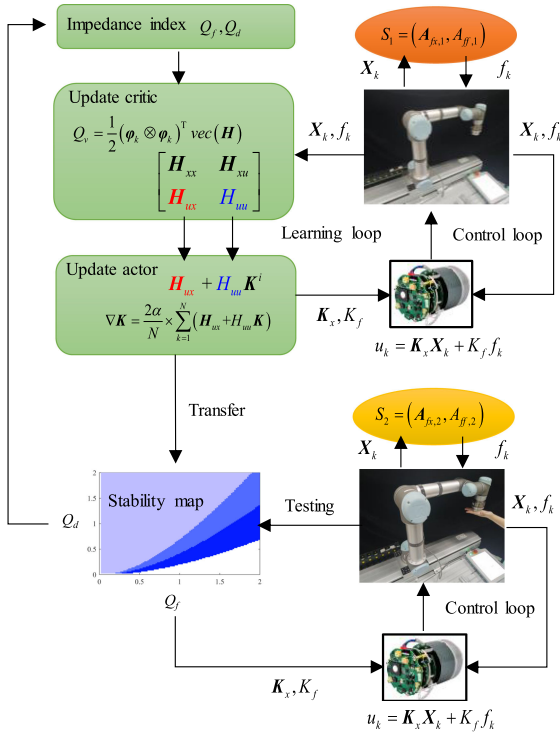


Fig. 2. Flowchart of the safety-learning strategy for impedance control.

After learning, an impedance control which is stable in S_1 is obtained. We have

$$\|\lambda (\mathbf{A}_d + \mathbf{B}_d \mathbf{K}_x)\|_2 < 1. \quad (33)$$

When the external force is related to the system state ($\mathbf{A}_{fx} \neq 0$), the criterion of the system stability is changed. In this case, the eigenvalue of the interactive system is

$$\lambda ((\mathbf{A}_d + \mathbf{B}_d \mathbf{K}_x) + (\mathbf{E}_d + \mathbf{B}_d K_f) \mathbf{A}_{fx}) \quad (34)$$

which is unnecessarily stable.

To find the boundary of the stability region, the maximum permissible external stiffness at which the system remains stable can be calculated. $\mathbf{A}_{fx}$ can be expressed as $[k_{\text{ext}} \ d_{\text{ext}}]$, where k_{ext} is the external stiffness and d_{ext} is the external damping. The maximum permissible external stiffness can be calculated by

$$\begin{aligned} & \max(k_{\text{ext}}) \\ & \text{with } \|\lambda ((\mathbf{A}_d + \mathbf{B}_d \mathbf{K}_x) + (\mathbf{E}_d + \mathbf{B}_d K_f) [k_{\text{ext}} \ 0])\|_2 < 1 \end{aligned} \quad (35)$$

where $\max(k_{\text{ext}})$ indicates the maximum permissible stiffness in the interaction environment. $\max(k_{\text{ext}})$ can be considered an index to show the robust region of the system. If the learned result has large maximum permissible stiffness, it can be easily transferred to the interaction environment and without introducing the stability issue.

However, in practice, the system parameters and external stiffness are unknown, and the maximum permissible stiffness cannot be directly calculated. Thus, a safety-learning strategy is presented to adjust the impedance index, as shown in Fig. 2.

Initial Q_f and Q_d are given, where Q_f depends on the desired impedance flexibility, and Q_d influences the response speed of the system. A model-based AC learning algorithm is then used to train the state and force feedback gains in training environment $S_1 = (\mathbf{A}_{fx,1}, \mathbf{A}_{ff,1})$. After training, the impedance control is tested in the testing environment $S_2 = (\mathbf{A}_{fx,2}, \mathbf{A}_{ff,2})$. If the controlled system is unstable in the testing environment, it is possible to increase Q_d or decrease Q_f to change the impedance property. Since Q_d and Q_f decouple the system convergence speed and the robot flexibility, the designer can easily adjust the impedance index according to the task.

V. SIMULATION RESULTS

The model-based AC learning of impedance control and the safety-learning strategy are investigated by the numerical simulation. In the simulation, a robot interacts with an external spring, as shown in Fig. 1. We assume that one joint (wrist joint) is controlled, and the acceleration of the joint is the input. The dynamic equation can be simplified to a second-order system as

$$\begin{aligned} \mathbf{X}_{k+1} &= \mathbf{A} \mathbf{X}_k + \mathbf{B} u_k + \mathbf{E} f_k, \\ \mathbf{A} &= \begin{bmatrix} 1 & 0.05 \\ 0 & 1 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ 0.05 \end{bmatrix}, \mathbf{E} = \begin{bmatrix} 0 \\ 0.25 \end{bmatrix}, \mathbf{X}_k = \begin{bmatrix} q_k \\ \dot{q}_k \end{bmatrix} \end{aligned} \quad (36)$$

where

$$\begin{cases} f_k = 0.1, \text{ in training} \\ f_k = k_{\text{ext}} (q_k + 0.1), \text{ in testing} \end{cases} \quad (37)$$

In training, a constant external force is given, i.e., $k_{\text{ext}} = 0$. After training, the controlled robot interacts with the testing environment, where k_{ext} is the stiffness of the testing environment. The impedance index is

$$Q_d = 0.25, Q_f = 1, R = 10^{-4}. \quad (38)$$

The control policy is

$$u_k = K_1 q_k + K_2 \dot{q}_k + K_f f_k = \mathbf{K} \bar{\mathbf{X}}_k. \quad (39)$$

To evaluate the performance of the learned result, the learned error is defined as

$$\eta_{\text{error}} = \|\mathbf{K} - \mathbf{K}_{\text{ARE}}\| \quad (40)$$

where $\mathbf{K}$ is the learned controller, and $\mathbf{K}_{\text{ARE}}$ is the controller solved using (10) with the given system parameters.

The learning process of model-based AC learning in the noise-free case is depicted in Fig. 3(a) and (b). Model-based AC learning has high learning efficiency, where the learned control can approach $\mathbf{K}_{\text{ARE}}$ within 100 s. When noise is added in the system (the variance of noise is 0.02), the learned results have a small deviation from $\mathbf{K}_{\text{ARE}}$, but the convergence process is stable, as shown in Fig. 3(c) and (d).

For comparison, the neural network-based AC learning algorithm is used to solve the same issue. The critic function is described by a two-layer neural network

$$\varphi_k = \sum_{i=1}^n \text{ReLU}(\mathbf{W}_{\varphi,i} [\bar{\mathbf{X}}_k \ u_k \ 1]^T) \quad (41)$$

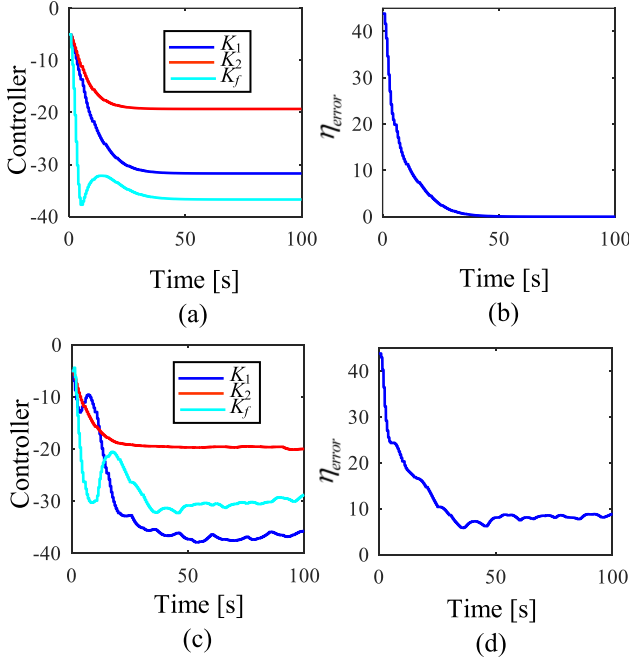


Fig. 3. Controller and the error index during learning when model-based AC learning is used. (a), (b) Noise-free. (c), (d) Variance of noise is 0.02.

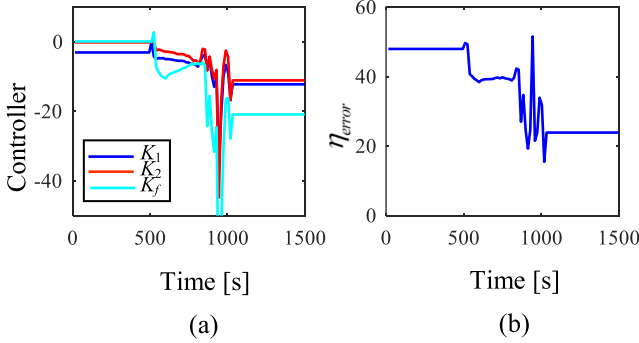


Fig. 4. (a) Controller and (b) error index during learning when neural network-based AC learning is used.

where $\text{ReLU}()$ is the rectified linear unit, and n is the number of hidden nodes, which equals 5. The update law of the neural network critic is

$$\varphi_{k+1} - \gamma\varphi_k \approx \frac{1}{2}\bar{\mathbf{X}}_k^T \bar{\mathbf{Q}} \bar{\mathbf{X}}_k + \frac{1}{2}u_k^T R u_k. \quad (42)$$

The actor is the same as (39)

$$u_k = \mathbf{K} \bar{\mathbf{X}}_k. \quad (43)$$

The update law of the actor is based on the deterministic policy gradient algorithm [27]

$$\mathbf{K}^{i+1} = \mathbf{K}^i + \alpha E_{u_k} [\nabla_{u_k} \varphi_k \nabla_{\mathbf{K}} u_k]. \quad (44)$$

The learning process of neural network-based AC learning in the noise-free case is depicted in Fig. 4. The learning time of neural network-based AC learning (about 1500 s) is much

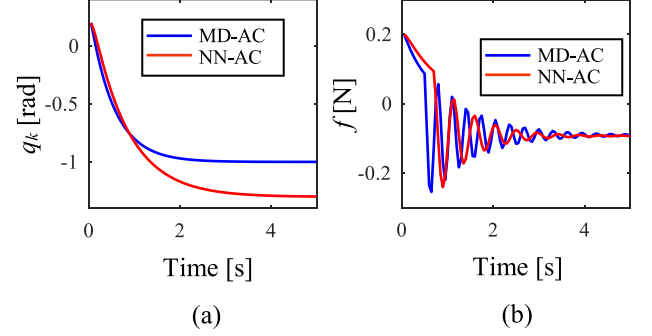


Fig. 5. Comparison results of model-based and neural network-based AC learning. (a) System response under 1N external force. (b) System response when it interacts with a spring $k_{\text{ext}} = 10\text{N/rad}$.

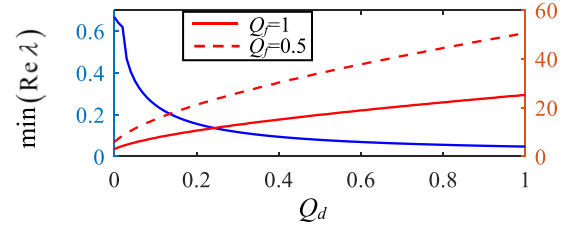


Fig. 6. $\min(\text{Re}\lambda)$ and $\max(k_{\text{ext}})$ with the increase of Q_d .

longer than the model-based AC learning algorithm since more unknown parameters exist. Also, the learned results deviate from $\mathbf{K}_{\text{ARE}}$ even in the noise-free case. The reason is that the neural network critic is an approximate function, and $\nabla_{u_k} \varphi_k = 0$ cannot always result $\mathbf{K} = \mathbf{K}_{\text{ARE}}$. By comparison, Theorem 1 shows that the equivalence of the model-based AC learning impedance control to $\mathbf{K}_{\text{ARE}}$. Fig. 5 shows the comparison results of model-based and neural network-based AC learning, where Fig. 5(a) shows the system response under 1N external force, and Fig. 5(b) shows the system response when it interacts with a spring $k_{\text{ext}} = 10\text{N/rad}$. Due to the deviation of the controller, the controlled systems show different responses. In model-based AC learning, the controlled system has the flexibility of 1 rad/N, which satisfies the desired flexibility.

The following simulations show the criterion to choose the impedance index. Fig. 6 shows the tendency of the minimum real part of eigenvalues and the maximum permissible external stiffness of the system with changed Q_d and Q_f . The left y-axis of Fig. 6 shows the minimum real part of eigenvalues, which indicates the convergence speed of the system. The right y-axis is the maximum permissible external stiffness when the controlled system interacts with an external spring. When Q_d increases, the maximum permissible external stiffness increases. The consequence is a low convergence speed of the original system. When Q_f decreases, the maximum permissible external stiffness increases, but the minimum real part of eigenvalues remains the same. Decreasing Q_f will not influence the convergence speed of the original system, but the flexibility of the robot will be reduced. The impedance index Q_f and Q_d are decoupled with each other.

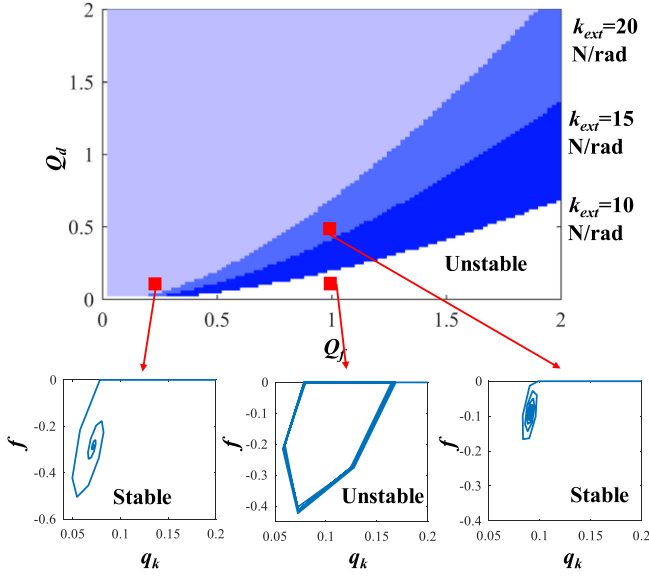


Fig. 7. Stability map of the interactive system.

The stability map of the system under model-based AC learning impedance control can be obtained due to the decoupling property of Q_f and Q_d , as shown in Fig. 7. The area of the stability map shows the stability region when the controlled system interacts with environments. The area decreases with the increase of the external stiffness. This map can be used as a guide to choose Q_f and Q_d . As Q_f and Q_d are weight factors of the cost function, they are not sensitive to the parameters of the original system. Also, the stability map is monotonic with Q_f and Q_d , which is friendly for designers. For example, if $Q_f = 1$ and $Q_d = 0.025$, the robot is unstable when it interacts with the external spring $k_{ext} = 10\text{N/rad}$. The designer can either reduce Q_f or enhance Q_d to stabilize the interactive system.

Remark 3: The criterion to transfer the learned control to a new environment is whether the system can stably interact with the new environment. When the system is unstable, Q_d or Q_f can be adjusted, and the system should be trained again. Given that Q_f and Q_d are decoupled, adjusting Q_d only influences the convergence speed of the system, whereas adjusting Q_f only influences the impedance flexibility. The decoupling characteristic helps the designer choose a suitable impedance index.

VI. EXPERIMENTAL RESULTS

In this section, experiments are conducted on a UR5 robot to verify the model-based AC learning algorithm. The robot system is shown in Fig. 8. A force sensor is set at the end of the robot arm. The robot is controlled using the robot operating system. The discount factor γ is 0.8, and the learning rate α is 0.01. The control period is set to 0.05 s. The maximum episode is set to 10 and each episode includes 10 s training time. Thus, the training can be finished within 100 s. A random exploration is required for RL. To ensure a safe training, a virtual constant force is provided in the force sensor to build the training environment.

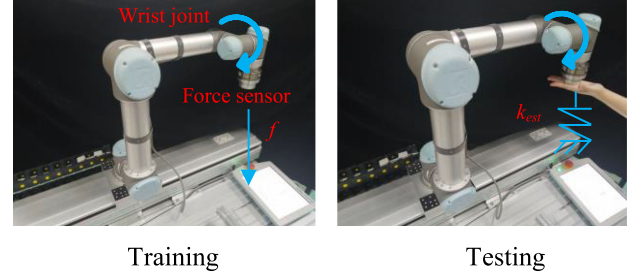
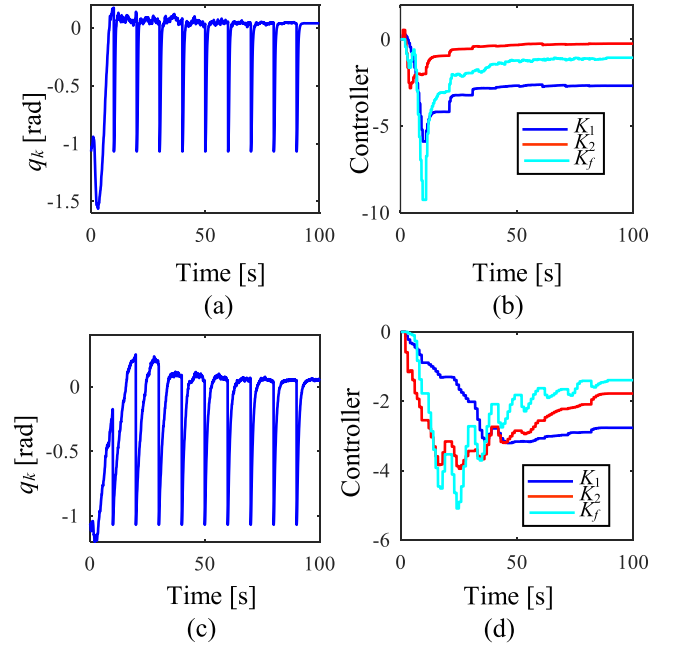


Fig. 8. Safety-learning strategy of impedance control for robot–human interaction.


 Fig. 9. State and the control gains during the learning process. (a), (b) $Q_d = 0.01$. (c), (d) $Q_d = 0.1$.

Case 1: human –robot interaction

Robots with impedance control can be used for the human–robot interaction. However, the stiffness of a human arm is unknown or even time-varying. Therefore, a safety-learning process is used to train the impedance control law of the robot. Initially, the impedance index is $Q_d = 0.01$, $Q_f = 1$, and $R = 10^{-4}$. Only the wrist joint is to be controlled.

The training process is shown in Fig. 9(a) and (b). After 100 s training, the impedance control law can be obtained. The impedance control input is

$$u_k = \mathbf{K}\bar{\mathbf{X}}_k, \quad \mathbf{K} = [K_1 \ K_2 \ K_f], \quad \bar{\mathbf{X}}_k = [q_k \ \dot{q}_k \ f_k]^T \quad (45)$$

where q_k is the wrist joint angle, $\dot{q}_k$ is the angular velocity, and f_k is the interaction force.

In the testing environment, no external force exists in the first episode (first 10 s). The system is stable and returns to the null position. In the second episode, a human palm resists

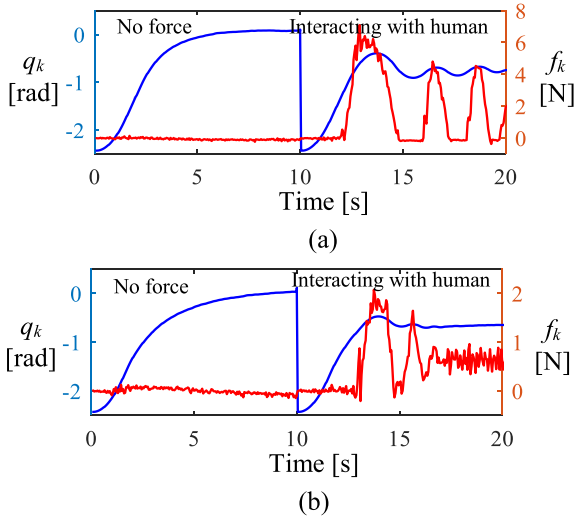


Fig. 10. Robot interacts with a human palm after learning. (a) $Q_d = 0.01$. (b) $Q_d = 0.1$.

the motion trail of the robot. When the robot interacts with the human palm, an unstable interaction exists, and the robot cannot stay in the human palm, as shown in Fig. 10(a). In the human–robot interaction scene, the flexibility of the robot Q_f should not change, such that the human can tutor the motion of the robot with the same force. Therefore, Q_d is enhanced to 0.1 to increase the stability region. The robot is trained again under the new impedance index, as shown in Fig. 9(c) and (d). With the increased Q_d , the robot has a low convergence speed but has a stable response when it interacts with the human palm. The flexibility of the robot is unchanged, as shown in Fig. 10(b). It should be noticed that the stiffness of the human palm is not constant. At the beginning of the human–robot collision, the stiffness of the human palm is large. The stiffness is reduced when the human is familiar with the interaction. This is the reason that a large interaction force is generated in the first interaction scene.

Case 2: Robot Machining

Robots with impedance control can be used for robot machining. During machining, a robot works like a spring and adapts to the workpiece surface to provide a constant machining force. The workpiece is a fixed aluminum alloy plate, and its stiffness is unknown. To find the suitable impedance property of the machining robot, the safety-learning strategy is used to learn the impedance control. The flexibility of the robot highly influences the machining quality. A high flexibility of the robot may introduce vibration since a force-induced vibration may be excited due to the vibration of the grinding head. A low flexibility of the robot may result in a large grinding force and damages the workpiece.

The robot machining system is illustrated in Fig. 11. The shoulder, elbow, and wrist joints are controlled. A grinding head is added to the end-effector of the robot. The wrist joint is tested with different Q_f to show the comparison results in terms of

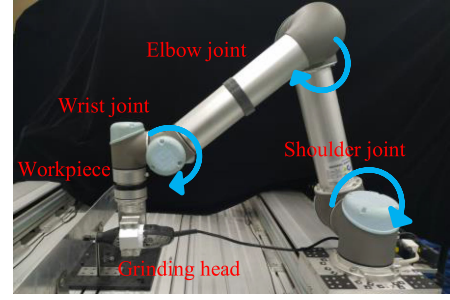


Fig. 11. Impedance control for robot machining.

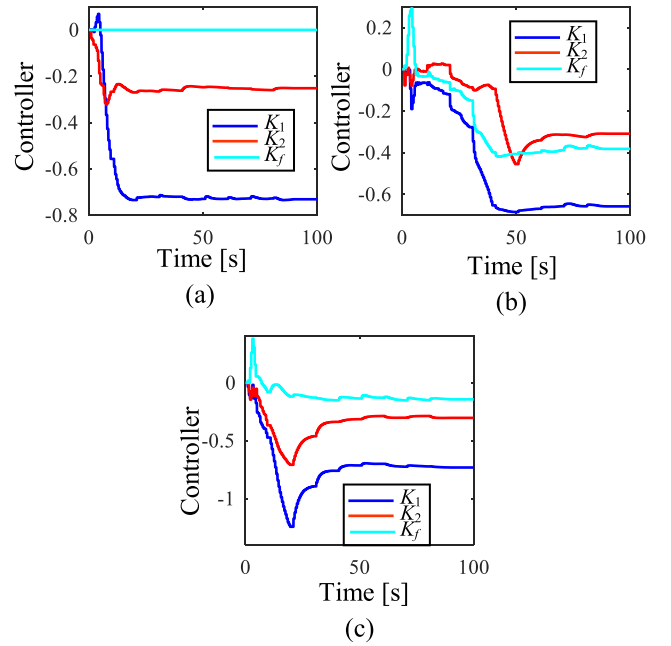


Fig. 12. Control gain during learning. (a) $Q_f = 0$. (b) $Q_f = 0.5$. (c) $Q_f = 0.2$.

$Q_f = 0$, $Q_f = 0.5$, and $Q_f = 0.2$. When the flexibility of the robot is zero, the robot is under position control. Large Q_f means high flexibility. Other parameters are the same ($Q_d = 0.1$ and $R = 10^{-4}$). The flexibility of the shoulder and elbow joint is set to zero ($Q_f = 0$) because those joints are far from the interaction position. The training process is depicted in Fig. 12. The impedance control of the robot can be obtained after 100 s training.

After learning, robot machining experiments are conducted. The control input of each joint during robot machining is

$$u_k = \mathbf{K}_x (\mathbf{X}_k - \mathbf{X}_{r,k}) + K_f f_k, \quad \mathbf{K}_x = [K_1 \ K_2], \mathbf{X}_k = [x_{1k} \ x_{2k}]^T, \mathbf{X}_{r,k} = [x_{1r,k} \ x_{2r,k}]^T \quad (46)$$

where $\mathbf{X}_{r,k}$ is the desired trajectory of each joint.

Fig. 13 shows the angle of the wrist joint and the corresponding grinding force. When $Q_f = 0$, the robot is under the position control. In this case, a large grinding force is introduced, and the average grinding force is approximately 40 N. The large grinding force results in the damage of the workpiece surface,

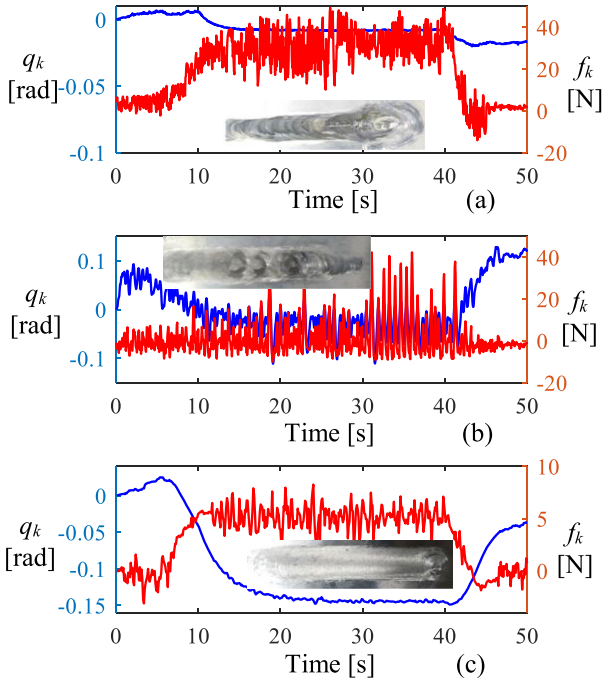


Fig. 13. Wrist joint angle and the interaction force as well as the surface quality of the workpiece. (a) $Q_f = 0$. (b) $Q_f = 0.5$. (c) $Q_f = 0.2$.

as shown in Fig. 13(a). When $Q_f = 0.5$, the robot has high flexibility. The vibration of the grinding hand can induce large vibrations of the robot arm. When it interacts with the workpiece, the vibration is reinforced. As a result, the grinding force shows large oscillation, and the surface texture of the workpiece is uneven, as shown in Fig. 13(b). When $Q_f = 0.2$, the robot has a suitable flexibility. An almost constant grinding force can be obtained, and a uniform surface quality is achieved, as shown in Fig. 13(c).

In the human–robot interaction and robot machining tasks, the learning environment is distinguished with the testing environment. The impedance control can be learned in safe environment and used in complex interaction environment. The experimental results verify the feasibility of model-based AC learning and the safety-learning strategy for the impedance control of robots. Designers can easily adjust the impedance index, and the robot can realize a stable interaction with the complex environment.

VII. CONCLUSION

In this article, a model-based AC learning algorithm was proposed to learn the optimal impedance control of robots. The impedance control was formulated into an optimization problem, and the impedance of the system can be adjusted using an impedance index. The model-based AC learning algorithm was used to learn impedance control. Given a model constraint, the learning process is rapid and reliable. The learned control gain is equivalent to the optimal impedance control gain solved using ARE. A safety-learning strategy was also proposed. The learning process is under a constant force, but the learned control can be further transferred to different interaction environments.

Two experiments were conducted, in terms of human–robot interaction and robot machining. The experimental results indicate that the robot can achieve a good interaction property after 100 s training.

Currently, model-based AC learning is efficient for linear or linearized systems. In the future, the extension of model-based AC learning to nonlinear systems will be studied.

VII. APPENDIX

Proof of Theorem 1 Substituting (15) into (16), the critic function can be expressed as

$$\begin{aligned} Q_v(\bar{\mathbf{X}}_k, \mathbf{u}_k) &= c_k + \gamma V(\bar{\mathbf{X}}_{k+1}) \\ &= \frac{1}{2}(\bar{\mathbf{X}}_k^T \bar{\mathbf{Q}} \bar{\mathbf{X}}_k + \mathbf{u}_k^T R \mathbf{u}_k) + \frac{1}{2} \gamma (\bar{\mathbf{X}}_{k+1} \otimes \bar{\mathbf{X}}_{k+1})^T \text{vec}(\mathbf{P}). \end{aligned} \quad (\text{A1})$$

Since $\bar{\mathbf{X}}_{k+1} = \bar{\mathbf{A}} \bar{\mathbf{X}}_k + \bar{\mathbf{B}} \mathbf{u}_k$, (a1) can be further expressed as

$$\begin{aligned} Q_v(\bar{\mathbf{X}}_k, \mathbf{u}_k) &= \frac{1}{2} \bar{\mathbf{X}}_k^T (\bar{\mathbf{Q}} + \gamma^2 \bar{\mathbf{A}}^T \mathbf{P} \bar{\mathbf{A}}) \bar{\mathbf{X}}_k + \frac{1}{2} \gamma \bar{\mathbf{X}}_k^T \bar{\mathbf{A}}^T \mathbf{P} \bar{\mathbf{B}} \mathbf{u}_k \\ &+ \frac{1}{2} \gamma \mathbf{u}_k^T \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{A}} \bar{\mathbf{X}}_k + \frac{1}{2} \mathbf{u}_k^T (R + \gamma \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{B}}) \mathbf{u}_k. \end{aligned} \quad (\text{A2})$$

The critic can be written in compact form as

$$\begin{aligned} Q_v(\bar{\mathbf{X}}_k, \mathbf{u}_k) &\approx \frac{1}{2} (\varphi_k \otimes \varphi_k)^T \text{vec}(\mathbf{H}), \\ \text{with } \varphi_k &= \begin{bmatrix} \bar{\mathbf{X}}_k \\ \mathbf{u}_k \end{bmatrix}, \mathbf{H} = \begin{bmatrix} \mathbf{H}_{xx} & \mathbf{H}_{xu} \\ \mathbf{H}_{ux} & H_{uu} \end{bmatrix} \end{aligned} \quad (\text{A3})$$

with

$$\begin{aligned} \mathbf{H}_{xx} &= \bar{\mathbf{Q}} + \gamma^2 \bar{\mathbf{A}}^T \mathbf{P} \bar{\mathbf{A}}, \mathbf{H}_{xu} = \gamma \bar{\mathbf{A}}^T \mathbf{P} \bar{\mathbf{B}}, \\ \mathbf{H}_{ux} &= \gamma \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{A}}, H_{uu} = R + \gamma \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{B}}. \end{aligned}$$

On the basis of the actor update law in (28), the steady actor is achieved when

$$\frac{2\alpha}{N} \sum_{k=1}^N (\mathbf{H}_{ux} + H_{uu} \mathbf{K}) = 0. \quad (\text{A4})$$

Substituting (a3) into (a4), we have

$$\mathbf{K} = -H_{uu}^{-1} \mathbf{H}_{ux} = - (R + \gamma \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{B}})^{-1} \gamma \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{A}}. \quad (\text{A5})$$

Then, substituting the actor (a5) into the Bellman equation (14), one obtains

$$\frac{1}{2} \bar{\mathbf{X}}_k^T \mathbf{P} \bar{\mathbf{X}}_k = \frac{1}{2} \bar{\mathbf{X}}_k^T \bar{\mathbf{Q}} \bar{\mathbf{X}}_k + \frac{1}{2} \mathbf{u}_k^T R \mathbf{u}_k + \frac{1}{2} \gamma \bar{\mathbf{X}}_{k+1}^T \mathbf{P} \bar{\mathbf{X}}_{k+1}. \quad (\text{A6})$$

Combining (a5) and (a6) yields

$$\bar{\mathbf{Q}} - \mathbf{P} + \gamma \bar{\mathbf{A}}^T \mathbf{P} \bar{\mathbf{A}} - \gamma^2 \bar{\mathbf{A}}^T \mathbf{P} \bar{\mathbf{B}} (R + \gamma \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{B}})^{-1} \gamma \bar{\mathbf{B}}^T \mathbf{P} \bar{\mathbf{A}} = 0. \quad (\text{A7})$$

When $\gamma = 1$, (a7) is equivalent to (10), and the policy (a5) is equivalent to (9). The equivalence between the policy from RL and that from the ARE is proved.

The following part shows that the learned control minimizes the defined value function (14) over all stabilizing controls. Note that

$$\frac{1}{2} \bar{\mathbf{X}}_k^T \mathbf{P} \bar{\mathbf{X}}_k = \frac{1}{2} \sum_{i=k}^{\infty} [\bar{\mathbf{X}}_i^T \mathbf{P} \bar{\mathbf{X}}_i - \bar{\mathbf{X}}_{i+1}^T \mathbf{P} \bar{\mathbf{X}}_{i+1}] + \frac{1}{2} \bar{\mathbf{X}}_{\infty}^T \mathbf{P} \bar{\mathbf{X}}_{\infty}. \quad (\text{A8})$$

Since $\bar{\mathbf{X}}_\infty^T \mathbf{P} \bar{\mathbf{X}}_\infty = 0$, we have

$$\begin{aligned} \frac{1}{2} \bar{\mathbf{X}}_k^T \mathbf{P} \bar{\mathbf{X}}_k &= \frac{1}{2} \sum_{i=k}^{\infty} [\bar{\mathbf{X}}_i^T \mathbf{P} \bar{\mathbf{X}}_i - \bar{\mathbf{X}}_{i+1}^T \mathbf{P} \bar{\mathbf{X}}_{i+1}] \\ &= -\frac{1}{2} \sum_{i=k}^{\infty} \left[\bar{\mathbf{X}}_i^T \left(-\bar{\mathbf{Q}} + \gamma^2 \bar{\mathbf{A}}^T \mathbf{P} \mathbf{B} (R + \gamma \mathbf{B}^T \mathbf{P} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{P} \bar{\mathbf{A}} \right) \bar{\mathbf{X}}_i \right. \\ &\quad \left. + \gamma \bar{\mathbf{X}}_i^T \bar{\mathbf{A}}^T \mathbf{P} \mathbf{B} u_i + \gamma u_i^T \mathbf{B}^T \mathbf{P} \bar{\mathbf{A}} \bar{\mathbf{X}}_i + \gamma u_i^T \mathbf{B}^T \mathbf{P} \mathbf{B} u_i \right]. \end{aligned} \quad (\text{A9})$$

Substituting (a9) into the value function (15), we have

$$\begin{aligned} V(\bar{\mathbf{X}}_k) &= \left[\frac{1}{2} \sum_{i=k}^{\infty} \gamma^{i-k} (\bar{\mathbf{X}}_i^T \bar{\mathbf{Q}} \bar{\mathbf{X}}_i + u_i^T R u_i) \right] \\ &\quad - \frac{1}{2} \bar{\mathbf{X}}_k^T \mathbf{P} \bar{\mathbf{X}}_k + \frac{1}{2} \bar{\mathbf{X}}_k^T \mathbf{P} \bar{\mathbf{X}}_k = \frac{1}{2} \bar{\mathbf{X}}_k^T \mathbf{P} \bar{\mathbf{X}}_k \\ &\quad + \frac{1}{2} \sum_{i=k}^{\infty} \gamma^{i-k} \left[\gamma^2 \bar{\mathbf{X}}_i^T \left(\bar{\mathbf{A}}^T \mathbf{P} \mathbf{B} (R + \gamma \mathbf{B}^T \mathbf{P} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{P} \bar{\mathbf{A}} \right) \bar{\mathbf{X}}_i \right. \\ &\quad \left. + \gamma \bar{\mathbf{X}}_i^T \bar{\mathbf{A}}^T \mathbf{P} \mathbf{B} u_i + \gamma u_i^T \mathbf{B}^T \mathbf{P} \bar{\mathbf{A}} \bar{\mathbf{X}}_i + u_i^T (R + \gamma \mathbf{B}^T \mathbf{P} \mathbf{B}) u_i \right] \end{aligned} \quad (\text{A10})$$

which can be written as

$$\begin{aligned} V(\bar{\mathbf{X}}_k) &= \frac{1}{2} \bar{\mathbf{X}}_k^T \mathbf{P} \bar{\mathbf{X}}_k + \frac{1}{2} \sum_{i=k}^{\infty} \gamma^{i-k} \\ &\quad \left[u_i + (R + \gamma \mathbf{B}^T \mathbf{P} \mathbf{B})^{-1} \gamma \mathbf{B}^T \mathbf{P} \bar{\mathbf{A}} \bar{\mathbf{X}}_i \right]^T \\ &\quad \times (R + \gamma \mathbf{B}^T \mathbf{P} \mathbf{B}) \left[u_i + (R + \gamma \mathbf{B}^T \mathbf{P} \mathbf{B})^{-1} \gamma \mathbf{B}^T \mathbf{P} \bar{\mathbf{A}} \bar{\mathbf{X}}_i \right]. \end{aligned} \quad (\text{A11})$$

Equation (a11) achieves its minimum when $u_k = -(R + \gamma \mathbf{B}^T \mathbf{P} \mathbf{B})^{-1} \gamma \mathbf{B}^T \mathbf{P} \bar{\mathbf{A}} \bar{\mathbf{X}}_k$, which is equal to the steady actor in (9) when $\gamma = 1$. Consequently, the learned control minimizes the value function (14). This completes the proof. $\square$

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Xingwei Zhao received B.S. and M.S. degrees from the University of Duisburg-Essen, Duisburg, Germany, in 2012 and 2013, respectively, and the Ph.D. degree from the Technical University of Berlin, Berlin, Germany, in 2017, all in mechanical engineering.

He is currently an Associate Research Fellow with the State Key Laboratory of Digital Manufacturing Equipment and Technology, Huazhong University of Science and Technology (HUST), Wuhan, China. His research interests mainly include nonlinear dynamics, robot control, and robotic manufacture.



technologies.

Shibo Han received the B.Eng. and M.Eng. degrees in mechanical engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2018 and 2021, respectively. He is currently working toward the Ph.D. degree in mechanical engineering with the Department of Mechanical Engineering, National University of Singapore, Singapore.

His research interests include distributed control, multiagent systems, and intelligent robot



Bo Tao (Member, IEEE) received the B.S. and Ph.D. degrees in mechanical engineering from the Huazhong University of Science and Technology (HUST), Wuhan, China, in 1999 and 2007, respectively.

After being a Postdoctor from 2007 to 2009, he has been an Associate Professor in 2009 and a Professor in 2013 with the School of Mechanical Science and Engineering, HUST, and he is currently a Changjiang Distinguished Professor with HUST. He has authored or coau-

thored more than 80 papers in international journals. His research interests include intelligent manufacturing and robotics technologies.



Zhouping Yin (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in mechanical engineering from the Huazhong University of Science and Technology (HUST), Wuhan, China, in 1994, 1996, and 2000, respectively.

He is currently a Professor with the School of Mechanical Science and Engineering, HUST. He has been a “Cheung Kong” Chair Professor with HUST since 2009. His research interests include electronic manufacturing equipment and technology, RFID technology and applications, and digital manufacturing and applications.

Dr. Yin was awarded the China National Funds for Distinguished Young Scientists in 2006.



Han Ding (Senior Member, IEEE) received the Ph.D. degree in mechanical engineering from the Huazhong University of Science and Technology (HUST), Wuhan, China, in 1989.

Supported by the Alexander von Humboldt Foundation, he was with the University of Stuttgart, Stuttgart, Germany, from 1993 to 1994. He was with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, from 1994 to 1996. He has been a Professor with HUST since 1997. He

is a “Cheung Kong” Chair Professor with Shanghai Jiao Tong University, Shanghai, China. His research interests include robotics, multiaxis machining, and equipment automation.

Dr. Ding was elected as a Member of the Chinese Academy of Sciences in 2013.